

Verifying Tartaglia's Theory of Cannonball Motion

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Abstract goes here

I. INTRODUCTION

Nicola Tartaglia was an Italian Renaissance mathematician, who was tasked with modeling the motion of cannonballs by the Doge of Venice in order to confer a military advantage. His predicted trajectory of a cannonball is shown in Figure 1.

[inset figure]

He also predicted the ranges for cannonballs shot at different angles, as shown in table 1. In this experiment, we aim to test his theories and see if they match predictions made by modern theories of physics.

Consider the horizontal and vertical components of projectile motion as given by Equations 1 and 2 where v_0 represents the initial velocity of the projectile, θ represents the initial angle, and x_0 and y_0 represent the initial positions on their respective axis.

$$x(t) = v_0 \cos \theta t + x_0 \quad (1)$$

$$y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2 - y_0 \quad (2)$$

In order to calculate the horizontal range, we need to compute when $y(t) = 0$ via quadratic formula, and substitute the resulting t value into $x(t)$.

Doing so, we get our final equation for which we fitted the data to.

$$\text{displacement} = v_0 \cos \theta * \frac{-v_0 \sin \theta - \sqrt{v^2 \sin^2 \theta - 4(y_0)(-\frac{1}{2}g)}}{-g} \quad (3)$$

II. METHODS AND MATERIALS

Methods here

It is appropriate to put links to code or datasets; put long items in an appendix or provide online. For example, all code used to generate this example is provided at <https://github.com/devangel77b/jse-mechanics-template/>.

A. Use of subsections

Subsection example

III. RESULTS

Results go here

Figure example: Fig. 1. Note that the figure reference name must begin with **fig:** in order for the $\text{\LaTeX}$ cleveref macros to properly parse it.

An example of a table is Table I.

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IV. DISCUSSION

Discussion goes here

V. ACKNOWLEDGEMENTS

Acknowledgements and references go here

FIGURES



FIG. 1. This is a caption. Captions for figures go below the figure.

TABLES

x	$\bar{x}$
0	1
1	0

TABLE I. This is a table. Captions for tables go above the table.